JAN-C-372

30 July 1946

JOINT ARMY-NAVY SPECIFICATION

CLEANER, RIFLE BORE

Army Number	Navy Number
2-117A	51C59

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specifications:

U. S. Army	Navy Department
2–117	Bureau of Ordnance
22 Feb. 1944	O. S. 1426
	11 Jan. 1944

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS.

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATION

JAN–P–124—Packaging and Packing for Overseas Shipment, Containers (Cans, Pails, and Drums) Metal.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[1]

FEDERAL SPECIFICATIONS

RR–S–866—Sieves; Standard, Testing.

VV–L–791—Lubricants and Liquid Fuels, General Specifications. (Methods for Sampling and Testing).

A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[1]

B. GRADE.

B–1. This specification covers one grade of rifle bore cleaner as hereinafter specified.

C. MATERIAL AND WORKMANSHIP.

C–1. The rifle bore cleaner shall be an organic soap emulsion or solution of water, paraffin oil and solvents so selected and compounded as to meet the requirements of this specification.

C–2. The rifle bore cleaner shall contain no materials which can have a harmful effect on personnel either by inhalation or contact under the conditions of its customary employment.

[1] Applicable only to Navy purchases.

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D. GENERAL REQUIREMENTS.

D–1. See section E.

E. DETAIL REQUIREMENTS.

E–1. *Flash point.*—Minimum, 100°F.

E–2. *Viscosity.*—At –20°F., maximum 2,000 centistokes.
At 100°F., 6–15 centistokes.

E–3. *Stability.*—When tested as specified in paragraph F–4b, the fluid shall show no separation.

E–4. *Performance.*—When applied in the manner specified in paragraph F–4c, the fluid shall clean test panels fired at by corrosive primers and shall prevent rusting.

E–5. *Alkalinity.*—The fluid shall be alkaline to litmus paper.

E–6. *Appearance.*—The fluid shall be clear and transparent when viewed by transmitted light.

E–7. *Residue.*—The residue left after evaporation at 130°F., shall be free from solid particles and shall not be tacky.

E–8. Odor.—The fluid shall be free from disagreeable and offensive odors.

E–9. *Low temperature fluidity.*—The fluid shall show movement in the test jar within 1 second or less when tested in accordance with paragraph F–4e.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Size of lot.*—

F–1a. *At place of manufacture.*—For the purpose of sampling, a lot of rifle bore cleaner shall consist of a manufacturers batch unless otherwise specified in the contract or order. If the material cannot be identified by batch, a lot shall consist of not more than 1,000 gallons offered for delivery at one time. (See par. H–3.)

F–1b. *At place of delivery.*—For the purpose of sampling, a lot of rifle bore cleaner shall consist of all the material in any one shipment.

F–2. *Sampling.*—Unless otherwise specified, sampling shall be conducted as follows: The Government inspector shall take a representative composite sample of not less than 1 quart of rifle bore cleaner from each lot in accordance with method 800.1 of Federal Specification VV–L–791 and the following specific procedures:

F–2a. *While filling containers at the place of manufacture.*—The Government inspector shall draw a sample at the discharge pipe where it enters the receiving vessel or vessels. The total sample shall be not less than 1 quart and shall be a composite of small samples of not more than 1 pint each, taken at regular intervals during the entire period of filling.

F–2b. *Filled containers.*—When the rifle bore cleaner is delivered in 1 gallon or smaller size cans, the sample shall consist of portions drawn from at least five containers.

F–3. *Inspection.*—

F–3a. *Army.*—Inspection shall be made as specified in the contract or order.

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F–3b. *Navy.*—All samples shall be forwarded to the Naval Gun Factory, Washington, D. C. for inspection and tests.

F–4. *Tests.*—Tests shall be made in accordance with the following paragraphs:

F–4a. The following tests shall be conducted in accordance with the requirements of Federal Specification VV–L–791.

Test	*Method No.*
Flash point	110.2
Viscosity	30.5

F–4b. *Stability.*—

F–4b(1). *Freezing.*—A sample of the fluid in a test tube shall be cooled to –65°F. for 1 hour and then allowed to warm to room temperature (70° to 80°F.). This freezing test shall be repeated for a total of six times. At the conclusion of this test, the sample shall be examined for separation. Following this procedure the sample shall be inverted once and allowed to stand at room temperature for 24 hours and again examined for separation.

F–4b(2). *Heating.*—A sample of the fluid in a loosely-stoppered test tube shall be heated for 30 minutes at 150°F., allowed to cool without agitation, and examined for evidence of separation.

F–4c. *Performance.*—

F–4c(1). *Preparation of test panels.*—Test panels, approximately 2 inches by 3 inches by 1/16 inch with well rounded edges, shall be prepared from open hearth low carbon steel. The panels shall be sandblasted to a fresh uniformly abraded surface using sandblasting sand which is practically white in color, sharp, dry and free from organic matter. The fineness of the sand shall be such that all shall pass a U. S. Standard No. 10 sieve, not more than 10 percent shall fail to pass a U. S. Standard No. 20 sieve and not more than 10 percent shall pass a U. S. Standard No. 50 sieve. Sieves shall conform to the requirements of Federal Specification RR–S–366. Badly rusted panels shall not be used and any oily residue on the panels shall be removed by solvent prior to sandblasting. Following sandblasting the panels shall be freed of dust either by rapping the edges against a clean object or by blowing with a jet of dried, filtered air. Immediately after cleaning, the panels shall be placed in a desiccator (calcium chloride) until used and shall be used the same day as prepared. The panels shall be handled with forceps at all times after the completion of the sandblasting operation in order to avoid difficulty with fingerprint corrosion.

F–4c(2). *Treatment of panels.*—Two test panels prepared in accordance with paragraph F–4c(1) shall have a primer (of the corrosive chlorate type) assembled in a caliber .30 ball ammunition cartridge (without powder charge or bullet) fired at the center of each. The cartridge shall be held in a rigid manner at 45° obliquity with a distance of 1 inch between the cartridge and the test plate. The panels shall then be cleaned with a cleaning patch saturated with the cleaning fluid using gentle swabbing. Follow this with a dry patch to remove all solution and dissolved salts. Next swab each panel with a clean patch saturated with the

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fluid. Place both panels in a humidity cabinet. The cabinet shall be maintained at 100 percent relative humidity and at 100°F. ± 2°F. for 3 days. The specified temperature and humidity conditions in the cabinet shall be maintained with fresh saturated air directed against the baffle plate to provide from 1 to 1.5 complete changes of air per hour. Auxiliary heat shall be applied through a distilled water layer maintained in the bottom of the cabinet. (A description of the cabinet used at Rock Island Arsenal is given in par. H–4.) During the test, there shall be definite evidence of moisture condensation on the test panels. The wood block supports holding the test panel shall be held in the cabinet in such position that the panels are parallel to the direction of the air flow in order that no panel shall act as a buffer for the panel behind.

F–4d. *Residue.*—Heat 25 ml of rifle bore cleaner in a flat-bottom Petri-type dish, approximately 10 cm in diameter, for 40 hours in an oven maintained at 130° ± 2°F. At the end of the test, the sample shall be cooled and examined for tackiness by touching the residue. Visual examination shall be made for solid particles.

F–4e. *Low temperature fluidity.*—A sample of the fluid shall be prepared in accordance with the procedure for pour point as described in method 20.1 of Federal Specification VV–L–791. The sample shall be held at –20°F. for 24 hours after which it shall be removed from the low temperature bath and the test jar tilted to ascertain whether movement occurs.

F–4f. *Alkalinity.*—A sample of the fluid shall be tested with litmus paper to determine compliance with the requirement of paragraph E–5.

F–4g. *Appearance.*—A sample of the fluid shall be viewed by transmitted light to determine compliance with the requirements of paragraph E–6.

F–4h. *Odor.*—A sample of the fluid shall be inspected for compliance with the requirements of paragraph E–8.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packaging (for domestic and overseas shipment).*—Unless otherwise specified, rifle bore cleaner shall be furnished in 2-ounce, 6-ounce, 1-quart, and 1-gallon screw top cans as specified in the contract or order. (See par. H–1.) Cans shall conform to the requirements of Joint Army-Navy Specification JAN–P–124.

G–2. *Packing (for domestic and overseas shipment).*—Unless otherwise specified, rifle bore cleaner shall be packed for shipment in accordance with the requirements of appendix I to Joint Army-Navy Specification JAN–P–124.

G–3. *Marking.*—Unless otherwise specified, individual containers and packing cases shall be plainly marked with the following information completed.

Material .....
Specification .....
Contractor .....
Net volume of contents .....

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Date of filling .....
Contract or order No. .....
Batch No. .....

In addition, shipments for the Navy shall be marked in accordance with the Navy Shipment Marking Handbook. The stencilling shall be protected by a coat of clear varnish to insure identification after extended storage.

H. NOTES.

H–1. Requests, requisitions, schedules, and contracts or orders should specify the following features:

(a) Title, number and date of the specificaton.

(b) Whether material is to be packaged, packed, and shipments marked for domestic or overseas shipment. (See pars. G–1 and G–2.)

(c) Kind of container in which delivery is desired. (See par. G–1.)

H–2. *Use.*—Cleaner, rifle, bore is a highly penetrating, mobile liquid which is used primarily for cleaning the bores of small arms, barrels, and artillery tubes and which forms a rust resisting coating for the cleaned surface.

H–3. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

H–4. A suitable humidity cabinet is a wooden box with the following dimensions:

Width 2 ft.
Height 2 ft.
Length 4-1/2 ft.

This box is constructed of approximately 1-1/2 inch white pine and is insulated with two 1/2 inch layers of celotex or equivalent. The interior is protected with two coats of bakelite varnish and one coat of high melting point petrolatum. Fresh saturated air passes through the cabinet at a rate of 1 to 1-1/2 changes per hour. This air enters the humidity cabinet at the middle of one end of the humidity cabinet behind a 7 inch diameter copper baffle plate held approximately 3 inches from the end wall. The air leaves the opposite end of the cabinet through a hole behind a similar baffle. The saturated air is prepared by passing air through a flow meter to the bottom of the saturating tank. This tank consists of a copper cylinder 5 inches in diameter and 24 inches high, which is equipped with a 500-watt immersion heater. Distilled water is maintained approximately 20 inches deep by means of an inverted 5 gallon bottle suitably connected to the saturating tank. This water is controlled at approximately 120°F. The desired temperature and humidity is maintained in the humidity cabinet by means of an additional immersion heater which is located in a pan of distilled water in the bottom of the cabinet. This pan is approximately 18 inches by 42 inches by 2-1/2 inches deep. Temperature control for the humidity cabinet regulates the heat supplied to this pan of water.

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H–5. Federal specification test methods referred to herein conform technically to ASTM methods of test as follows:

Test	Federal Specification Method No.	ASTM No.
Flash point	110.2	D-93
Pour point	20.1	D-97

H–6. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained, as indicated in the "Index of United States Army, Joint Army–Navy, and Federal Specifications Used by the War Department". Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–7. Copies of Joint Army-Navy specifications and Federal specification (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–8. Copies of this Joint Army–Navy specification (required for Army purchases) may be obtained from the office, Chief of Ordnance, War Department, Washington D. C.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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